

Impact-Aware State Estimation for Off-Road Motion

Problem

- Off-road: Shocks, slip, rapid attitude changes
- IMU EKF: Fixed process noise
→ Drift & Failures



Approach

IMU-Only EKF



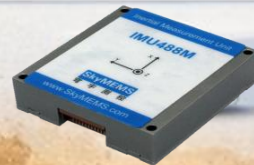
Fixed Q

Impact-Aware EKF + LSTM



Adaptive $\alpha_x Q$

- Accuracy: ATE / RPE
- Robustness: High-Jerk Segments?



IMU + Camera (VIO)



Plan & Timeline

- ✓ Week 1: IMU EKF Baseline
- ✓ Week 2: Jerk-Based Heuristic
- ✓ Week 3: EKF + LSTM Integration
- ✓ Week 4+: (Stretch) VIO Extension?



Pranav:
EKF Baselines & Eval.



Avantika:
LSTM Training & Integration